

1 Different Aspects of Abstractness Predict Vocabulary Acquisition Across Languages

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Abstract

of vocabulary data from 87507 children

Introduction

Young children rapidly learn new words, but some words are slower to enter children's vocabularies than others. For example, infrequent words are challenging for children to learn [CITE], as the child has few exposures from which to draw a robust mapping between the word and the referent [CITE]. Abstract words are the quintessential "hard words" (Gleitman, Cassidy, Nappa, Papafragou, & Trueswell, 2005), but the explanatory mechanism is murkier, in part because abstractness encompasses several interrelated lexical properties. In a recent consensus paper, abstractness was defined as: > "(1) (historical): Referring to the quality of a concept (or word meaning) that has no sensory or motor salience (in opposition to concrete) in that it cannot be seen, heard, touched, felt, smelled, tasted or acted upon. (2) (contemporary): The quality of a concept (or word) whose meaning is understood primarily on the basis of language, but also draws from interoceptive experiences, including emotion, introspection, and metacognition. Abstract concepts are often exemplified by perceptually dissimilar associated actions and events." (Reilly et al., 2023).

Within this single definition, we have multiple, non-exclusive, potential challenges for word learning. 1) Abstract words can be characterized by an absence of sensory or motor experience with the referent (e.g., want, Paivio, 1971, 1991). 2) Abstract words afford fewer opportunities for iconic mappings, leaving their phonological forms less informative. 3) Abstract words may involve interoception, emotion, introspection, metacognition, or the characterization of relationships between entities. 4) Abstract words may be less contextually specific (Davis, Altmann, & Yee, 2020; Hoffman, Lambon Ralph, & Rogers,

2013; e.g., problem, Schwanenflugel & Shoben, 1983), and so are exemplified by ”
 perceptually dissimilar associated actions and events.” (Reilly et al., 2023). For example,
 unlike words like “ball” or “dog”, there is no single situation that clearly teaches what
 “thing” means¹.

It is fairly uncontroversial to state that children learn abstract words later (Bergelson
 & Swingley, 2013; Della Rosa, Catricalà, Vigliocco, & Cappa, 2010; Muraki, Reggin,
 Feddema, & Pexman, 2025; Ponari, Norbury, Rotaru, Lenci, & Vigliocco, 2018), but why
 exactly they learn abstract words later is open to debate. The most oft-cited reason why
 children tend to acquire abstract words later than concrete ones is because abstract words
 lack a clear sensory-perceptual referent, and thus are characterized by referential
 uncertainty (e.g., Bergelson & Swingley, 2013). However, this explanation rests on a
 negative definition of abstract words—namely, whereas concrete words refer to things that
 can be experienced directly through the senses, abstract words refer to things that cannot,
 and must be instead defined by other words (Brysbaert, Warriner, & Kuperman, 2014).
 This view on abstract words is increasingly coming into question, with a growing literature
 pointing to the richness of experiences associated with abstract words, emphasizing not
 only that understanding abstract words tends to rely more on language associations
 (Borghi et al., 2019; Paivio, 1991) and that they tend to be more ambiguous and deployed
 in diverse lexical and situational contexts (Barsalou, Dutriaux, & Scheepers, 2018; Davis et
 al., 2020; Hoffman et al., 2013; Schwanenflugel & Shoben, 1983) but also that they tend to
 be highly associated with emotional experience (Kousta, Vigliocco, Vinson, Andrews, &
 Del Campo, 2011) and social knowledge (Wilson-Mendenhall, Simmons, Martin, &
 Barsalou, 2013). There are also differences between abstract and concrete words in
 iconicity (i.e., the degree to which a word’s meaning is reflected in its form (Hinojosa,
 Haro, Magallares, Duñabeitia, & Ferré, 2021; Winter, Lupyan, Perry, Dingemanse, &

¹ Philadelphian readers who have tried to explain “jawn” to an out-of-towner are likely familiar with this particular challenge.

Perlman, 2023), such that abstract words counterintuitively tend to be rated as more iconic. Taking a multidimensional perspective on abstract words enables a more precise question: which specific properties of abstractness actually make a word harder to learn?

In the following sections, we consider four different ways in which a word meaning might be abstract: 1) a lack of associated perceptual experience, 2) social-emotional, cognitive, or interoceptive meaning, 3) iconic vs. arbitrary mappings between word form and meaning, and 4) broad contextual diversity. These dimensions are theoretically separable, differentially distributed across the lexica of various languages, and potentially associated with distinct learning mechanisms.

Different Constructions of Abstractness

Abstractness as absent of sensorimotor experience. Embodied theories propose that sensorimotor processes are essential for learning the meanings of words [e.g., CITE]. Under this view, children should learn words more easily when their referents afford rich, distinctive sensorimotor experiences. Developmental evidence is consistent with the idea that children’s early vocabularies reflect their perceptual experiences. For example, English-speaking sighted children, but not blind children, were more likely to produce words when the words are more strongly associated with visual experience (Campbell, Casillas, & Bergelson, 2024). Richer sensory experience generally supports earlier word learning; Seidl, Indarjit, and Borovsky (2023) observed that English words tend to be learned earlier when their semantic associations span multiple senses. In related experimental work, Seidl et al. (2023) observed that 2-year-olds learn words more readily when they are able to interact with the referents than when they are only able to see the referents. These behavioral findings align with neuroimaging work showing that relative to concrete words, abstract words elicit reduced activation in cortical regions supporting sensory perception (Wang, Conder, Blitzer, & Shinkareva, 2010), consistent with the idea that perceptual grounding facilitates processing.

Motor experience may provide an additional source of grounding beyond sensory associations: Words that afford rich opportunities for physical engagement [e.g., xxx,xxx, as captured by body-object-interaction BOI ratings; cite; cite] are acquired earlier and processed more efficiently in English (Inkster, Wellsby, Lloyd, & Pexman, 2016; Pexman, 2019; Suggate & Stoeger, 2017; Thill & Twomey, 2016; Wellsby & Pexman, 2014). These effects also relate to children’s own motor development: the processing speed of high-BOI words, but not low-BOI words, is linked to children’s fine motor skills (Suggate & Stoeger, 2017), suggesting that developing motor abilities may support access to action-related meanings. Converging evidence from adults shows that adults learn novel words better when the words are with relevant actions (Casasanto & de Bruin, 2019; Vukovic & Shtyrov, 2019), and that motor cortical regions are engaged during the acquisition of motor-relevant vocabulary (Liuzzi et al., 2010; Vukovic & Shtyrov, 2019). Because infants and toddlers across cultures differ substantially in the timing and opportunities for motor exploration (Lohaus et al., 2014), the contribution of motor grounding to word learning may also vary across languages and developmental contexts.

Effects of sensorimotor grounding may differ across sensorimotor modalities and across development. Visual, tactile, auditory, and motor experiences provide distinct types of information about a referent, and some modalities may offer more reliable cues for early learners than others (San Roque, Norcliffe, & Majid, 2024). These contributions may also shift with age: sensorimotor experience may be especially important before infants can infer meanings from language alone. On the other hand, effects of sensorimotor associations could grow stronger across age, as infants are increasingly mobile and able to explore their environments.

Abstractness as wordform arbitrariness. A second source of difficulty for abstract words involves the limited opportunities they afford for transparent, non-arbitrary mappings between a word’s phonological form and its meaning. Iconicity refers to a perceptually-motivated resemblance between form and meaning (e.g., the English word

moo resembles the sound a cow makes). Work on onomatopoeias, ideophones, and systematicity show that iconicity is a more prevalent phenomenon than previously recognized [cite], but languages vary widely in the prevalence and strength of iconicity in their lexicons (with iconicity especially prevalent in sign languages and in many spoken languages, including several Niger-Congo languages, Quechua, Japanese, Korean, Turkish, and Basque, Akita & Pardeshi, 2019). Sensorimotor aspects of word meaning afford different kinds of iconic mappings across languages: in spoken languages, auditory meanings are more likely to be expressed iconically, whereas in sign languages, visual and tactile meanings are more likely to be expressed iconically (B. Thompson, Perlman, Lupyan, Sehyr, & Emmorey, 2020; Winter et al., 2023). Because abstract words lack strong sensorimotor grounding, they typically afford fewer opportunities for such motivated mappings, and their phonological forms may therefore provide learners with less information about word meaning (though c.f., Hinojosa et al., 2021; Winter et al., 2023).

Across languages, words that are higher in iconicity tend to be learned earlier (Laing, 2019; Massaro & Perlman, 2017; Perry, Perlman, & Lupyan, 2015; Sidhu, Williamson, Slavova, & Pexman, 2021). One account attributes this advantage to easier referential mapping between the referent and the wordform (Perniss, Özyürek, & Morgan, 2015; R. L. Thompson, Vinson, Woll, & Vigliocco, 2012). Indeed, computational and experimental evidence from preschoolers and adults suggest that iconic words are easier to acquire than non-iconic words (Imai, Kita, Nagumo, & Okada, 2008; K. F. Kantartzis, 2011; K. Kantartzis, Imai, Evans, & Kita, 2019; Miyazaki, Imai, Kantartzis, Kita, & Kita, 2013; Nygaard, Cook, & Namy, 2009; Yoshida, 2003, 2012), and it has been theorized that acquiring iconic words early scaffolds the later acquisition of non-iconic words (Imai & Kita, 2014; Imai et al., 2008; R. L. Thompson, 2011; though c.f., Nielsen & Dingemanse, 2021). Other accounts, however, challenge the assumption that young children possess the world knowledge necessary to detect and abstract perceptual resemblance. Perhaps instead, effects of iconicity are actually driven by frequency or salience in child-directed

speech (Laing, Keren-Portnoy, Khattab, & Sloggett, n.d.; Perry, Perlman, Winter, Massaro, & Lupyan, 2018), types of meanings typically encoded by iconic words (B. Thompson et al., 2020; Winter et al., 2023), or systematicity (Campbell et al., 2025). Studying effects of iconicity across languages allows us to start to disentangle iconicity from its lexical covariates.

Abstractness as internal or relational meanings. Many abstract words are associated with emotional/affective (Kousta et al., 2011; Troche, Crutch, & Reilly, 2017; Troche, Crutch, & Reilly, 2014; Villani, Lugli, Liuzza, & Borghi, 2019), interoceptive (Connell, Lynott, & Banks, 2018; Harpaintner, Trumpp, & Kiefer, 2018; Lynott, Connell, Brysbaert, Brand, & Carney, 2020), social/relational (Harpaintner et al., 2018; Troche et al., 2017; Villani et al., 2019; Wiemer-Hastings & Xu, 2005), and/or unobservable cognitive states (Harpaintner et al., 2018; Troche et al., 2017, 2014). We group these facts of abstractness together here based on vibes (our conceptual intuition) and on the principal component analysis of abstract words by Troche et al. (2017).

Abstract words tend to relate more strongly to emotional experience than concrete words (Lenci, Lebani, & Passaro, 2018; Newcombe, Campbell, Siakaluk, & Pexman, 2012; Villani et al., 2019; Winter, 2022). According to the Affective Embodiment Account [cite cite], for words without consistent perceptual mappings, emotional relevance may support processing and acquisition (Desai, Reilly, & van Dam, 2018; Goriachun, Strijkers, Gala, & Ziegler, 2025; Kousta et al., 2011; Newcombe et al., 2012; Ponari, Norbury, & Vigliocco, 2017; Wild & Kuperman, 2024). Developmental and psycholinguistic evidence supports this proposal. In English, emotionally valenced words (both positive and negatively valenced) tend to be acquired earlier than neutral words (Muraki et al., 2025; Ponari et al., 2018, 2017; Reggin, Muraki, & Pexman, 2021; Wild & Kuperman, 2024), and this effect is especially strong for abstract concepts. For concrete concepts, by contrast, only positively valenced meanings show an early-acquisition advantage (Ponari et al., 2017). Children also show an advantage of valence in novel word learning tasks (Ponari, Norbury, & Vigliocco,

2020). In adults, strong emotional associations facilitate faster lexical processing across a variety of tasks (Pauligk, Kotz, & Kanske, 2019; Yao et al., 2018; Yao, Scott, Bruce, Monteith-Hodge, & Sereno, 2025), suggesting that emotional information may increase support the retrieval of word meanings. On the other hand, emotion words themselves are learned relatively late overall, and older children learn new emotion terms more readily than younger children (Shablack, Becker, & Lindquist, 2020). From the perspective of the Affective Embodiment Account, this suggests a nuanced picture: emotional experience can ground abstract meanings broadly and early, even though the lexicon of dedicated emotion labels only becomes richly differentiated later in childhood.

Interoception is a perceptual domain which relates to internal sensations (e.g., *hungry, sleepy, anxious*). Unlike other perceptual domains, however, interoception is more strongly associated with abstract concepts than concrete concepts (Connell et al., 2018). There is a strong link between interoceptive and emotional experience (Connell et al., 2018; Critchley & Harrison, 2013; Wiens, 2005), such that interoceptive signals shape both the phenomenological “feel” of an emotion and the conscious awareness of that feeling, with individual differences in interoception predicting variation in both (Schandry, 1981; Wiens, 2005). Accordingly, interoception is a defining feature of emotion concepts, particularly those tied to fear and sadness (Connell et al., 2018). Interoception could be part of the grounding mechanism for emotion concepts, set forth by the Affective Embodiment Account (Kousta et al., 2011). This prominence of interoceptive experience in emotion concepts is consistent with the Affective Embodiment Account (Borghi et al., 2017; Kousta et al., 2011; Ponari et al., 2018), which argues that emotional connotations help ground abstract meanings because many abstract words refer to internal, physiological states that reflect and reconstruct emotion. Interoceptive strength is likewise associated with earlier age of acquisition (Reggin et al., 2021).

Social relevance may also support abstract word processing and acquisition (Diveica, Muraki, Binney, & Pexman, 2024; Mahmood et al., 2025; though c.f., Goriachun et al.,

2025, who find mixed effects of socialness on processing tasks across languages). Following
 Diveica, Pexman, and Binney (2022), we use socialness to refer to the degree to which a
 word’s meaning encompasses socially relevant content, including social characteristics of a
 person or group (*friendly*), social behaviors or interactions (*share*), and social institutions
 (*school*). Words from infants early social routines (*hi*, *mommy*, *thank-you*, *byebye*) are
 among the earliest learned (C. Caselli, Casadio, & Bates, 1999; M. C. Caselli et al., 1995;
 Frank, Braginsky, Yurovsky, & Marchman, 2017; Gleason & Weintraub, 1976; Gopnik,
 1988; Schults, Tulviste, & Konstabel, 2012; Tardif et al., 2008), and across languages, these
 early social-routine words represent roughly a quarter of children’s first 10 words (Casey,
 Potter, Lew-Williams, & Wojcik, 2023; Frank, Braginsky, Yurovsky, & Marchman,
 2021). These social words are frequent in language input to children, yet unlike concrete
 nouns (which refer to the same referent ~80% of the time, Bergelson & Swingley, 2013;
 Custode & Tamis-LeMonda, 2020), words like “*bye*” only refer to the same situation ~48%
 of the time (Casey et al., 2023). This substantial variability in the perceptual information
 available accompanying these words (Casey et al., 2023; Clerkin, Hart, Rehg, Yu, & Smith,
 2017) could make it challenging for children to grasp word meaning. However, their
 appearance in predictable, high-frequency social routines may offer some compensatory
 stability that facilitates learning (Gleason, Perlmann, & Greif, 1984; Gleason & Weintraub,
 1976). According to the Words as Social Tools framework, abstract meanings may
 especially benefit from social interaction, which allows learners to negotiate shared
 meaning through verbal interactions (Borghi & Binkofski, 2014; Borghi, Mazzuca, &
 Tummolini, 2025). In this way, even with substantial referential ambiguity, social words
 may gain an advantage from the predictable, socially structured interactions in which they
 are used (Pexman, Diveica, & Binney, 2023).

Cognitive and mental-state terms could pose a distinct challenge for learners because
 they refer to unobservable internal states that require the representation of others’ beliefs,
 desires, or intentions. Mental-state words (e.g., *think*, *know*, *pretend*) typically enter

children’s speech between 2.5 and 5 years of age (Mulder, 2015; Pascual, Aguado, Sotillo, & Masdeu, 2008; Shatz, Wellman, & Silber, 1983; Wellman & Estes, 1987), a period during which many children already produce 250–600 words (Frank et al., 2017). One possible explanation for these difficulties is that the concepts underlying these words are hard to grasp, and therefore their acquisition must await relevant conceptual development. Although infants show early sensitivity to agents’ goals and preferences (with some emergent sensitivity as early as 3 months, Sommerville, Woodward, & Needham, 2005; Woodward, 2009), these capacities continue to mature throughout the preschool years (Rakoczy, 2022; Yott & Poulin-Dubois, 2016), and many children do not reliably pass false-belief tasks until around age four (Liu, Wellman, Tardif, & Sabbagh, 2008; Wellman, Cross, & Watson, 2001) (which itself is strongly correlated with children’s language skills, Milligan, Astington, & Dack, 2007). Mental-state information is not inherently salient for mapping onto words; even adults typically attend to actions rather than to agents’ internal states unless cued that a word refers to cognitive / mental states, and children are less sensitive to these cues than adults (Papafragou, Cassidy, & Gleitman, 2007). Across languages, at least in the domain of evidential morphology, children seem to be able to perform *non*-linguistic mental state reasoning tasks before they can understand / produce the corresponding mental state language, suggesting that the conceptual foundations precede language (Ozturk & Papafragou, 2016; Papafragou, Li, Choi, & Han, 2007). Pyers and Senghas (2009) studied a group of deaf learners of an emerging sign language in Nicaragua, and found that success on false-belief theory-of-mind task corresponded to the acquisition and production of mental state terms. Another explanation is that (at least, in the case of English mental state verbs) these words appear in are syntactically complex, and perhaps this structure is difficult for children to acquire. Yet, even controlling for the frequency and the syntactic contexts in which these words tend to appear, words referring to cognition or mental states are acquired later [cite cite]. Wellman & Silber (1983) suggest that the earliest uses of mental state verbs tend to be idiomatic or rote learned, with

phrases like *I don't know*, which may not require representation of mental states, just knowledge of the context in which those phrase appear. Similarly, children will produce markers of knowledge states before they show robust comprehension of knowledge states (Ünal & Papafragou, 2016), so production of cognitive words (or morphemes, as was tested by Ünal & Papafragou, 2016) may not wait for conceptual understanding. Ultimately, however, across the preschool years, children's uses of mental-state words become increasingly diverse and sophisticated across age² (Booth, Hall, Robison, & Kim, 1997; Pascual et al., 2008; Shatz et al., 1983; Wellman & Estes, 1987).

By hypothesis, then, the most difficult abstract words to acquire should be those that lack all of these potential grounding resources: words with limited sensorimotor experience, weak emotional arousal, and minimal social relevance (e.g., *category*, *quantity*, *reason*).

Word Meaning Abstract from Specific Circumstance. According to the Dual Coding Theory (Dove, 2009; Paivio, 1986, 1991, 2010), while concrete words have sensorimotor associations, abstract words are more reliant on their linguistic associations, and indeed, neuroimaging evidence points to a heightened role of the verbal systems for abstract words relative to concrete words (Jeffrey R. Binder, Desai, Graves, & Conant, 2009; J. R. Binder, Westbury, McKiernan, Possing, & Medler, 2005; Wang et al., 2010). While adults can readily learn abstract words from linguistic context (Mkrtychian et al., 2021), children's ability to do this may depend on their nascent language skills (T. C. Lund, Sidhu, & Pexman, 2019). As children build language skill, they can more readily extract meaning from syntactic structure (Babineau, de Carvalho, Trueswell, & Christophe, 2021; Babineau, Havron, Dautriche, de Carvalho, & Christophe, 2022; Gleitman, 1990) or from word's co-occurrence with other words (Huebner & Willits, 2018; Lany & Saffran, 2010; Lenci et al., 2018; K. Lund & Burgess, 1996; Syrett & Lidz, 2010; Wojcik & Saffran, 2013).

² Although rare, even children's youngest uses can show evidence of mental state representation: "*I thought* is was an alligator. Now *I know* it's a crocodile." (Wellman & Estes, 1987)

Context Availability Theory likewise proposes that abstract words necessarily depend more on their connection to the context in which they are presented (Schwanenflugel, Akin, & Luh, 1992), which can vary quite widely. Indeed, more contextually diverse words tend to be more abstract (Hoffman et al., 2013). Words like *idea* can occur in a broad range of situations, whereas concrete words such as *spaceship* appear in far narrower contextual ranges. This flexibility may help explain why contextual information plays a disproportionate role in abstract word learning and processing. A substantial body of work shows that the linguistic environments surrounding a word predict when that word is learned, but the direction of these effects is variable. Some studies find that contextual diversity impedes word learning: Words in distinctive linguistic contexts tend to be earlier learned in naturalistic work (Roy, Frank, Decamp, Miller, & Roy, 2015) and easier learned in experimental work [cite cite]. The diversity of contexts a word appears in predicts word learning in both children (Hills, Maouene, Riordan, & Smith, 2010; Kachergis, Yu, & Shiffrin, 2009; Rosa, Salom, & Perea, 2022) and adults (Johns, Dye, & Jones, 2016). These effects have been interpreted in line with two processes in early word learning: the lure of the associates, which suggests that words learned earlier tend to be better connected with a range of other known words (Hills, Maouene, Maouene, Sheya, & Smith, 2009); and preferential acquisition, where words learned earlier tend to be used in a range of distinct contexts (Hills et al., 2010). For example, in young adults contextual diversity facilitates different stages of learning, with low-diversity contexts aiding initial meaning formation and high-diversity contexts producing stronger, more context-independent representations (Mak, 2021).

Developmental Change in Mechanisms

It's possible that the mechanisms supporting acquisition of abstract words change across development. As children mature, their growing cognitive, social, and linguistic skills may pave the way for increasingly sophisticated word learning (Babineau et al., 2021;

Meylan & Bergelson, 2022). Particularly if we conceptualize abstract words as acquired primarily through linguistic context, as children’s linguistic comprehension continues to mature, we should expect to find that effects of linguistic context on AoA weaken across age. Similarly, if word learning is supported by sensorimotor experience with a word referent, as children’s motor abilities advance, and children are increasingly able to manipulate objects and independently locomote, we would expect to find sensorimotor effects strengthen across toddlerhood.

Crosslinguistic Variability in Mechanisms

The age of acquisition of words is correlated to varying degrees across languages (Łuniewska et al., 2016, 2019). Studying patterns of language acquisition across languages is informative for understanding which acquisition mechanisms are persistent across languages and cultures, versus which mechanisms are specific to a particular developmental or linguistic context. For example, work with Mandarin learners called into question the universality of the *noun bias* in children’s early vocabulary (Bornstein et al., 2004; Hardegger et al., 2024; Kauschke, Lee, & Pae, 2007; Tardif et al., 2008). The strength and direction of lexical effects on age of acquisition can vary across languages; for instance, while frequency and shorter word length consistently predict earlier word production across many languages, effects of emotional valence are more variable (Braginsky, Yurovsky, Marchman, & Frank, 2019). Additionally, within a given language, there are commonly strong correlations between lexical properties; e.g., in English but not ASL, auditoriness is correlated with frequency (See Appendix 1). Thus, it can be difficult to interpret whether an early-learned word is early-learned because of *Lexical Property X* or *Lexical Property Y* (Reilly & Kean, 2007). By looking across languages, we can utilize cross-linguistic variability in the distribution of lexical properties to achieve better power to isolate individual lexical properties.

The Present Study

The present study examines the cross-linguistic stability of associations between lexical properties traditionally linked to abstractness and age of acquisition. In doing so, we hope to uncover, across many languages: what is it about abstractness that makes a word challenging to learn? This study is only possible due to massive effort from researchers around the world to document lexical properties and vocabulary. We leverage lexical ratings collected by hundreds of researchers from thousands of participants. The vocabulary data presented here span tens of thousands of young children. Today's backdrop of open science practices makes such an analysis feasible.

Methods

CDI

the MacArthur-Bates Communicative Development Inventory (CDI), a vocabulary measure in which parents are asked to indicate whether their child understands and/or produces each of several hundred words Fenson et al. (1994). Since its conception, the CDI has been adapted into many languages, validated with many populations, and administered to many thousands of children.

When the CDI is adapted into a new language, researchers of each language tend to consult previous CDIs, parents, educators, and child development experts to curate a culturally-relevant set of early-learned words (Jarůšková, Smolík, Chládková, Oceláková, & Paillereau, 2023). Thus, the CDIs contain partially-overlapping subsets of words (e.g., words for mother and father are ubiquitous across CDI forms, but acorn, magpie, and paprika appear on only the Latvian, Beijing Mandarin, and Hungarian forms, respectively).

We pull data from Wordbank, a large, open database of CDI data (Frank et al., 2017), using the wordbankr R package (Braginsky, 2024). For the present analyses, we take

the full set of participants from Wordbank as of Wed Nov 26 20:56:43 2025. This results in a sample of $N = 87507$.

Lexical Properties

All lexical ratings are first re-scaled (within languages) to a 1 to 10 scale for better comparability across languages and lexical properties. For zipfian-distributed norms (e.g., corpus-based frequency), norms were first logged and then scaled.

To achieve maximum lexical rating coverage across words and languages, while recognizing that language-specific ratings by native speakers are likely the best source of lexical data, we developed the following rules for imputation:

- For norms pertaining to the underlying concept (e.g., concreteness, perceptual strength, emotional valence):
 - If norms are available for a specific word in a specific language, the scaled value will be used.
 - If norms are not available for a specific word or not available in a specific language, then the mean rating for the words' translation equivalent from all available languages will be used.
 - If norms are not available for a specific word or its translation equivalent, then any words without norms will be excluded from the analysis.
- For norms pertaining to the wordform (e.g., frequency, iconicity, phonological properties):
 - If norms are available for a specific word in a specific language, the scaled value will be used.

- If norms are not available for a specific word but are for other words in that language, then any words without norms will be excluded from the analysis.
- If there are no norms at all for a language along one of these scales, then any set of analyses using that norming scale will exclude that language.

https://docs.google.com/spreadsheets/d/1fivS85qayBJtp_nXjVmAvwZn7u6F517RFur0UPTJcAw/edit#gid=0

Imageability.

Perceptual properties. More recently, researchers have developed more specific measures of words’ perceptual associations. Beyond general perceptibility, these norms separate associations by modality (visual, tactile, auditory, haptic, gustatory, interoceptive), yielding the modality specific ratings, an overall strength rating, and a measure of the spread of ratings across modalities (Lynott et al., 2020). Such specificity allows researchers to test whether information from certain senses, like vision or touch, may be privileged over others [cite], as well as whether receiving sensory information from multiple modalities may support acquisition (e.g., Seidl et al., 2023). For example, the Lancaster Sensorimotor Norms, created for English, Using similar instructions, sensory norms have been created for other languages, including Chinese (Zhong, Wan, Ahrens, & Huang, 2022), Spanish (Díez-Álamo, Díez, Alonso, Vargas, & Fernandez, 2017), French (Miceli, Wauthia, Lefebvre, Ris, & Simoes Loureiro, 2021), Russian (Miklashevsky, 2018), and Italian (Vergallito, Petilli, & Marelli, 2020).

BOI. Body-object interaction (BOI) ratings measure the extent to which a human body can physically interact with a word’s referent (Siakaluk, Pexman, Aguilera, Owen, & Sears, 2008), and in particular, seem to index ease of graspability (Heard, Madan, Protzner, & Pexman, 2019). In neuroimaging studies, semantic processing of high BOI (vs low BOI) words is associated with neural regions involved in grasping (Hargreaves et al., 2012). In line with embodied theories of cognition, words’ BOI ratings predict the speed of

lexical processing in children and adults [CITE] as well as age of acquisition (Muraki, Siddiqui, & Pexman, 2022; Pexman, 2019; Thill & Twomey, 2016). Child-centered BOI ratings, gathered by asking parents of elementary-aged children “how easily the average 6-year-old can physically interact (using the body: hands, mouth, etc.) with what each word represents” (Muraki et al., 2022, p. pg.9), may capture this variability even better; though in English, child BOI ratings and adult BOI ratings correlate strongly ($r = 0.67$) with each other (Muraki et al., 2022).

Iconicity. At the time of publication, iconicity norms were only available for English (Winter et al., 2023), ASL (N. K. Caselli, Sehyr, Cohen-Goldberg, & Emmorey, 2017; Sehyr, Caselli, Cohen-Goldberg, & Emmorey, 2021), BSL, Japanese [CITE], and Spanish (Hinojosa et al., 2021). We did not attempt to interpolate values for this property given its intrinsic connection to phonological form, which varies cross-linguistically.

xxx.

Table 1

Lexical properties. Examples of words that are high/low on each property.

Property	Source	top_3	bottom_3	uni_lemma_coverage
Visual	Dutch [speed2022], English [lynott2020], Estonian [], French [miceli2021], Italian [vergallito2020], Mandarin [zhong2022], Russian [miklashevsky2018], Spanish [diez-alamo2017]	orange (10), naked (9.8), video (9.8)	sound (1.9), smell (1.8), smelly (1.8)	97%
Auditory	Dutch [speed2022], Estonian, French [miceli2021], Italian [vergallito2020], Mandarin [zhong2022], Russian [miklashevsky2018], Spanish	noise (9.6), speak (9.5), noisy (9.4)	purple (1.1), sweets (1), ribbon (1)	96%
Haptic	Dutch [speed2022], Estonian, French [miceli2021], Italian [vergallito2020], Mandarin [zhong2022], Russian [miklashevsky2018], Spanish	touch (9.8), smooth (9.3), cuddle (9.3)	listen (1), wanna (1), speak (1)	96%

Olfactory	Dutch [speed2022], Estonian, French [miceli2021], Italian [vergallito2020], Mandarin [zhong2022], Russian [miklashevsky2018], Spanish	smelly (9.7), pizza (8.8), smell (8.7)	flood (1), thump (1), divide (1)	96%
Gustatory	Dutch [speed2022], Estonian, French [miceli2021], Italian [vergallito2020], Mandarin [zhong2022], Russian [miklashevsky2018], Spanish	pizza (9.7), donut (9.6), bagel (9.5)	thump (1), grasp (1), divide (1)	96%
Interoceptive	Dutch [speed2022], English [lynott2020], French?? [miceli2021], Mandarin [zhong2022]	hungry (9.2), sleepy (8.8), happy (8.6)	letter (1), skate (1), drain (1)	96%
Emotional Arousal	Dutch [verheyen2020], English [mohammad2018], Estonian, French [CITE], Italian [], Mandarin [xu2022], Swedish [blomberg2015], Turkish [torkamani-azar2019; conca2024]	shoot (9.3), shout (9.3), burst (9)	snail (2), asleep (1.9), cotton (1.6)	88%
Imageability	Croatian [peti-stantic2021], Dutch [verheyen2020], French [desrocher2009], Italian, Mandarin [su2023], Norwegian, Portuguese [soares2017], Russian [miklashevsky2018], Spanish, Swedish [blomberg2015], Turkish [conca2024]	jacket (9.8), bleed (9.8), around (9.8)	which (3.6), gotta (3.6), leave (3.2)	76%
Concreteness	Croatian [peti-stantic2021], Dutch [verheyen2020], English [brysbaert2014], Estonian [], French [bonin2018], Italian, Portuguese [soares2017]	boots (10), camera (10), bagel (10)	until (1.7), though (1.4), would (1.3)	96%
Context Diversity	*all* [Wikipedia], except ASL, BSL, etc.	NA	NA	NA
Body Object Interaction	English [body], French [miceli2021], Spanish []	bread (9.9), boots (9.8), cereal (9.8)	alert (2.2), shame (2.1), whose (1.9)	77%

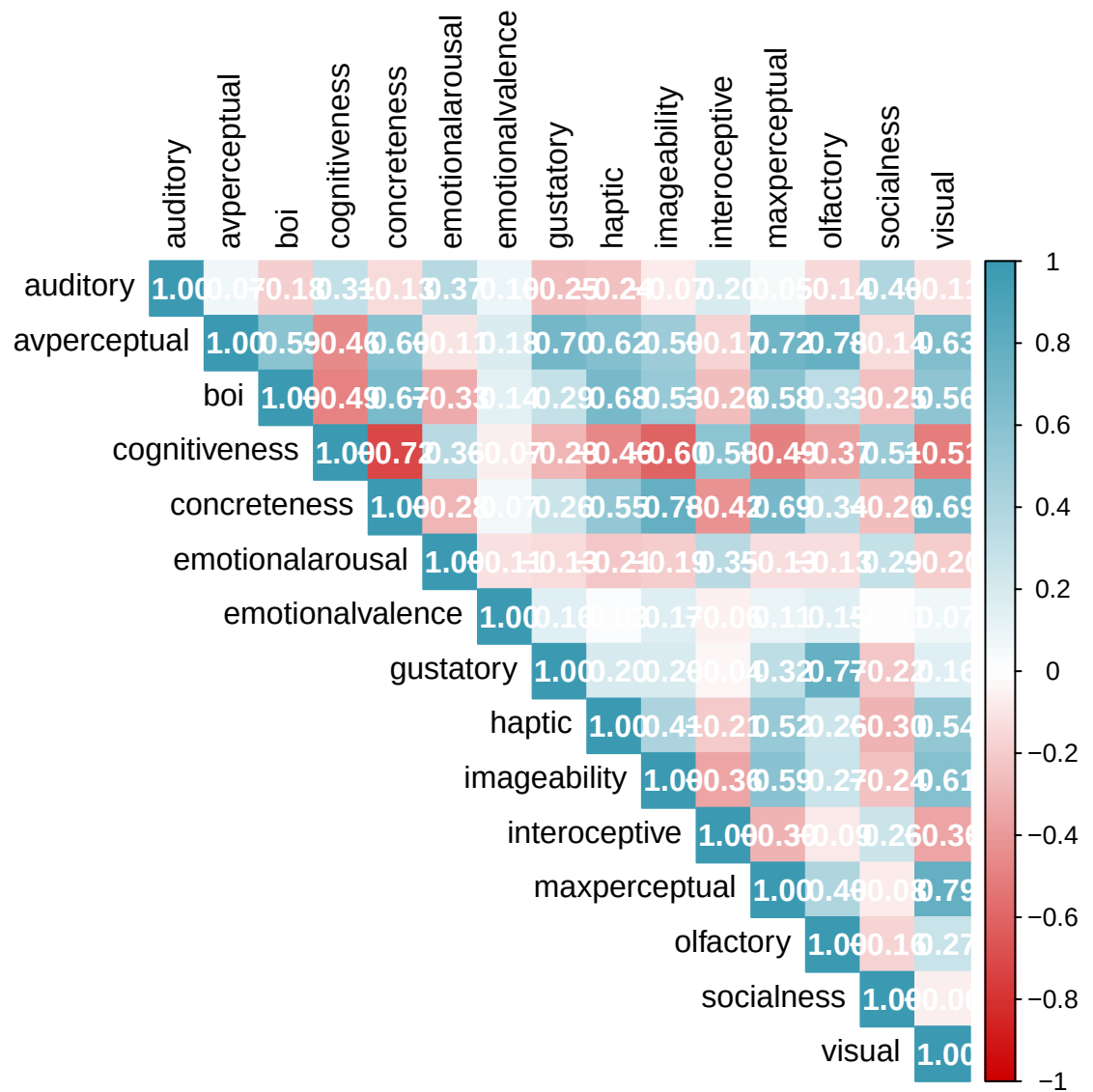
Iconicity	ASL [@caselli2017a], BSL [@ortega2025], English [@winter2023a], Spanish [@hinojosa2020], Japanese [CITE]	squeak (9.7), vroom (9.6), click (9.6)	below (2.2), could (2.1), really (1.8)	84%
Max Perceptual	xxx	orange (10), touch (9.8), naked (9.8)	which (2.3), rather (2.3), y'all (2.1)	97%
Average Perceptual	xxx	pizza (7.2), fruit (6.8), donut (6.8)	rather (1.8), wanna (1.7), don't (1.5)	96%
Word Frequency	*all* [Wikipedia], except ASL, BSL, etc.	NA	NA	NA
Socialness	diveica2024	people (9.9), family (9.6), cuddle (9.2)	yogurt (1), light (1), cloud (1)	52%
Emotional Valence	need to fill in	happy (10.4), tiger (10.4), police (9.9)	punish (2.3), wrong (2.2), thief (2)	41%
Cognitiveness	corenblum2025	smart (8.8), clever (8.8), brain (7.9)	eraser (1.1), donkey (1.1), cereal (1.1)	38%

Covariates. All models included the following variables as covariates: age (from Wordbank, rounded to the nearest month), frequency (from Wikipedia, or from subjective native speaker ratings), phonological complexity (for spoken languages, word length; for sign languages, xxx), lexical category (from Wordbank; levels included: nouns, predicates, function words, other)

Results

Relationships among Abstractness Variables

First, we computed pairwise Pearson's correlation coefficients for all variables, displayed in Figure 2, together with histograms showing the distribution of each variable and scatterplots showing how variables relate to each other. Here, we summarize across languages, but the by-language plots can be found in Supplementals (possibly as confusion matrix). Also the correlations for a given property across languages in supplement



Abstractness effects on word production

First, we fit a baseline statistical models for each language where word production is modeled as a combination of age, frequency, and phonological complexity. We then added each of our target predictors. For each target predictor for each language, it is thus possible to extract an effect size estimate of the target predictor and its interaction with age on the likelihood of word production. To explore heterogeneity and to estimate central tendency and variation in the effects, we borrow from meta-analytic techniques, treating each language as a study, and creating a weighted average of the effect size.

Figure 1 shows the coefficient estimate for each predictor in each language; see Table xxx for a full accounting of abstractness effect sizes. We find that iconicity is the strongest predictor of acquisition (mean across languages and measures: $B = 0.13$ (range: 0.03 - 0.43)). Many of the sensorimotor variables were relatively strong predictors of word production, including body-object-interaction ($B = 0.14$ (range: 0.06 - 0.43)), imageability ($B = 0.18$ (range: -0.25 - 0.46)), hapticness ($B = 0.09$ (range: 0 - 0.29)), and visualness ($B = 0.05$ (range: -0.06 - 0.37)). The effects of the social-emotional variables socialness ($B = -0.01$ (range: -0.41 - 0.12)) and emotional arousal ()—as well as several of the other sensorimotor variables (olfactoriness, interoceptiveness, auditoriness, and gustatoriness)—had much smaller mean effects on children’s word production. Interestingly, concreteness was not reliably associated with word production across languages ($B = 0.04$ (range: -0.08 - 0.2)). Finally, the effect of context diversity was highly variable, with some languages showing large positive effects (e.g.,), while others showed large negative effects. Looking across languages, words which were used in more contexts were less likely to be produced ($B = 0$ (range: -0.24 - 0.17)).

Developmental Change. Positive age interactions can be seen in at least xxx out of xxx languages for visualness, hapticness, body-object-interaction, and imageability. Conversely, there are negative age interactions for auditoryiness and iconicity in at least

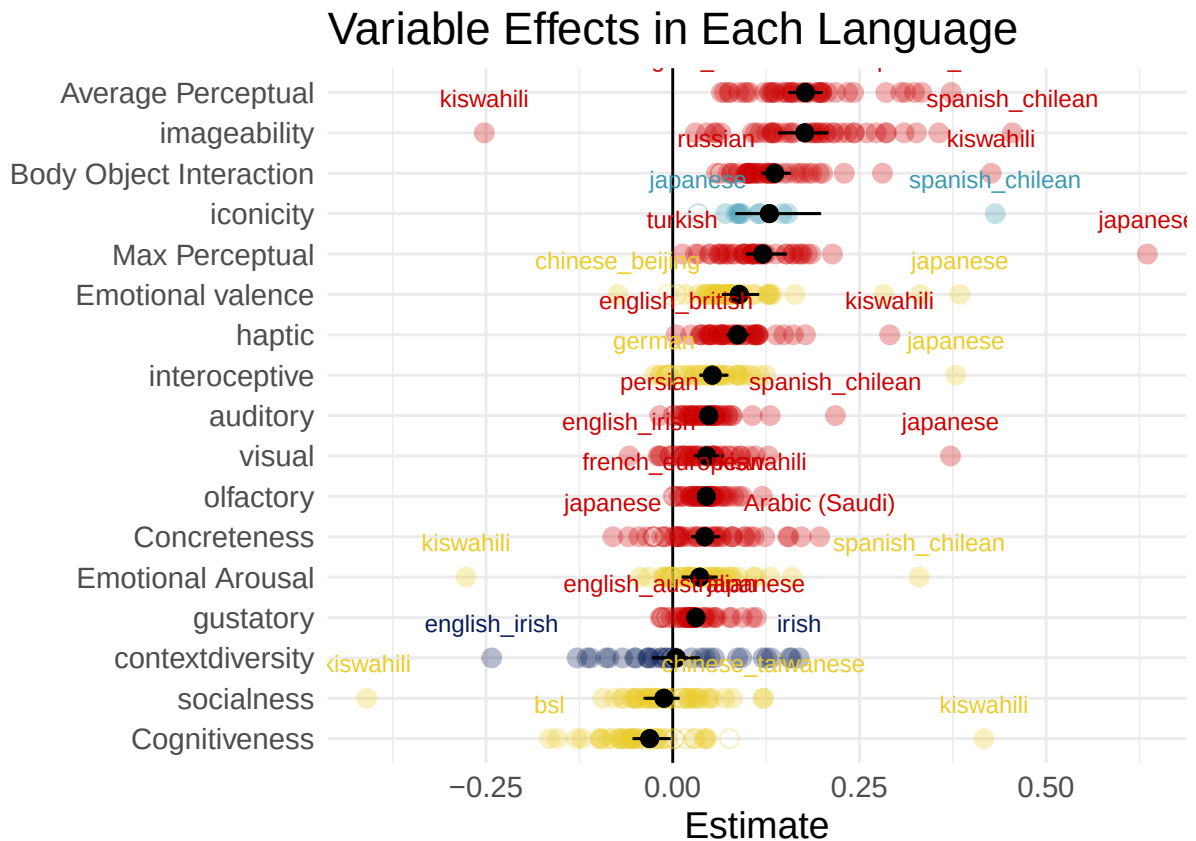


Figure 1. Effect size estimates (from logistic regression) of each of the abstractness variables studied. Each dot represents the effect size of one language, with filled-in dots showing significant effects and empty dots indicating that the effect was not significant in that language. Black dot and whiskers show mean effect size with standard error.

xxx out of xxx languages. This suggests that certain sensorimotor properties (chiefly, children's ability to see and touch a word's referent) facilitate learning more so later in development, while iconicity and auditoriness³ facilitate learning earlier in development. This result is consistent with the speculation above that xxx.

³ The interaction between auditoriness and age was near 0 for both American Sign Language ($B = \text{xxx}$) and British Sign Language ($B = \text{xxx}$).

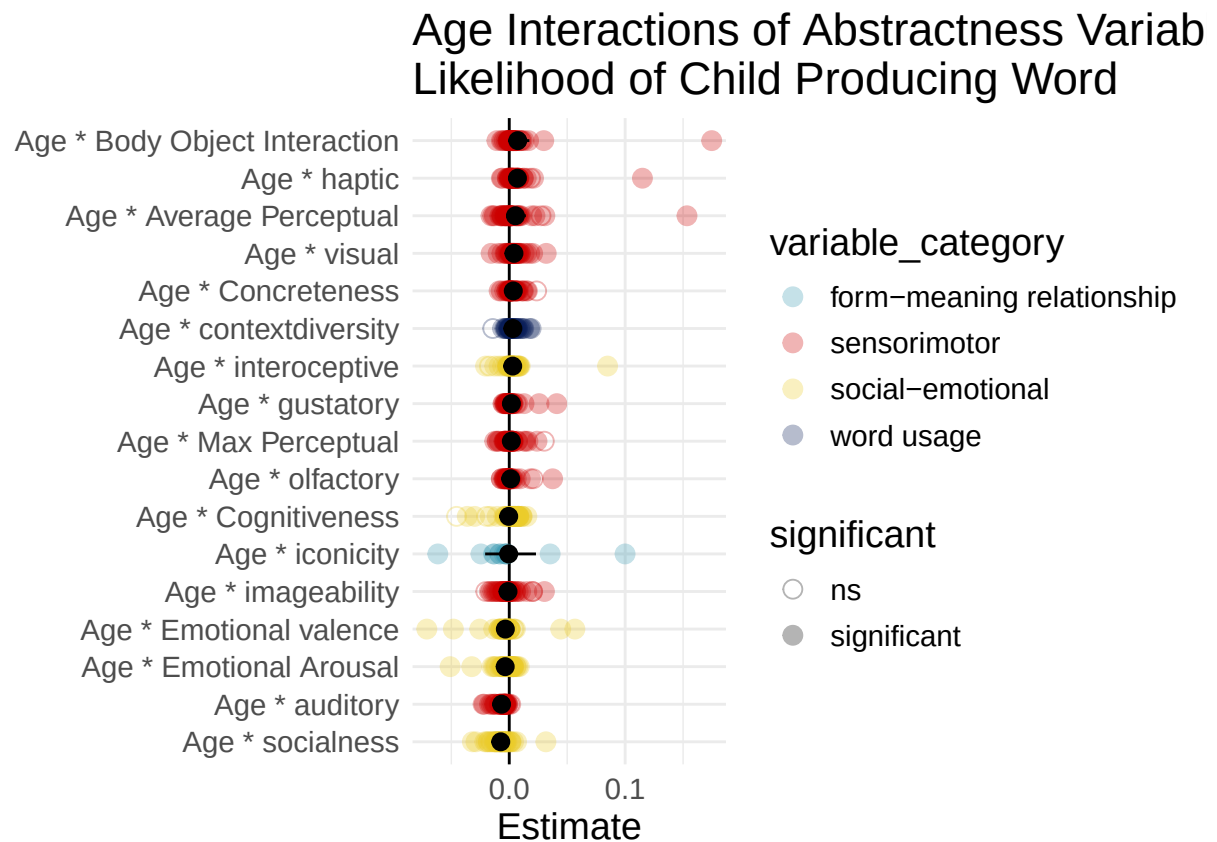


Figure 2. Effect size estimates (from logistic regression) of the interaction between each of the abstractness variables and age. Each dot represents the effect size of one language, with filled-in dots showing significant interactions and empty dots indicating that the effect was not significant in that language. Black dot and whiskers show mean effect size with standard error. Positive values show that effects are stronger for older children, while negative values show that effects are larger for younger children.

What drives abstractness effects?

Finally, we conduct an exploratory analysis aimed at measuring which of these abstractness variables best accounts for variability in words' production. Because these variables are interrelated, as shown in Figure xxx, when adding them to the same model, there is a risk of collinearity (Tomaschek et al., 2018). To get around this issue, we compare the predictive strength of the abstractness variables using random forest regression, implemented with the ranger package in R (Wright & Ziegler, 2017). Random Forest regression is a machine learning method based on decision trees and recursive partitioning. Each tree is fit to a subset of the data and only uses a subset of the predictors (a third of the available variables in our analyses). The predictions of many hundreds of trees can then be averaged, which helps avoid overfitting and increases accuracy. Because Random Forest regression does not need to estimate a coefficient for every predictor variable in every tree, this approach allows us to avoid issues of collinearity, but the relative importance of two highly correlated predictors can still be compared by assessing the trees in which they do not co-occur.

This approach yields variance importance scores for each of the variables in our model. In Figure XXX, we display the overall ranking of the variance importance scores. We find that XXX is most important, followed by xxx.

Importance plots from random forest regression

We next explore the variability in ranking across languages. intra-class correlation of importance by variable across languages. we find an ICC of $R = xxx$, indicating xxx stability in relative importance of these predictors across languages.

Multiple behavioral tasks - abstractness effects

picture naming lexical decision

Discussion

Parents may not have an accurate sense of their children's comprehension of emotion words (Roepstorff, Mayor, Havighurst, & Kartushina, 2024).

An additional proposal is that abstract meanings are grounded through conceptual metaphors, such as affection is warmth, which map abstract ideas onto more concrete sensory experiences (Lakoff, 2012; Lakoff & Johnson, 1980). Such metaphors may help children connect abstract words like love to tangible embodied experiences (e.g., the warmth of being held). However, metaphorical thinking develops relatively late—typically between 8 and 10 years of age (Dove, 2011; Winner, Rosenstiel, & Gardner, 1976), well after children have acquired many early abstract adjectives such as yucky and pretty (Frank et al., 2017; Reggin et al., 2021). Thus, while metaphorical mappings may shape adult representations, they are unlikely to account for the earliest abstract words in children's vocabularies.

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